Flutter App Business Proposal 2.0.1

End-to-End iOS, Android & Admin Panel Architecture using Flutter 3.x

1. EXECUTIVE SUMMARY & PROJECT VISION

This business proposal outlines the technical scope, milestone schedule, and commercial framework for designing, developing, testing, and deploying a high-performance cross-platform mobile application built with Google Flutter (Dart) for iOS and Android, backed by a scalable cloud REST/GraphQL API. Version 2.0.1 incorporates updated UI/UX component specifications, push notification workflows, and payment gateway compliance required for production release on Apple App Store & Google Play.

2. TECHNICAL ARCHITECTURE & DELIVERABLES

- **Frontend Mobile App:** Flutter 3.x (Dart), Bloc / Riverpod state management, native 60fps animations
- **Backend & API Layer:** Node.js / Laravel REST APIs, JWT OAuth2 security, Redis caching layer
- **Database & Storage:** PostgreSQL / MySQL cluster with AWS S3 media storage & CloudFront CDN
- **Third-Party Integrations:** Firebase Cloud Messaging (FCM), Razorpay / Stripe, Google Maps SDK
- **Web Admin Console:** Role-based dashboard for user management, analytics, and content moderation

3. HIGH-LEVEL MILESTONE & DELIVERY SCHEDULE

Phase	Deliverable Module	Duration	Sign-Off Criteria
Sprint 1	UI/UX Wireframes, Figma Prototype & Design System	2 Weeks	Design Approval
Sprint 2	Core Database Schema, Auth APIs & Flutter Shell	3 Weeks	Alpha Build
Sprint 3	Core Business Modules, Real-Time Sync & Notifications	3 Weeks	Beta Demo
Sprint 4	Payment Gateway, Admin Dashboard & QA Regression	2 Weeks	UAT Sign-off
Sprint 5	App Store & Google Play Store Submission & Handover	1 Week	Production Live

4. KEY HIGHLIGHTS OF VERSION 2.0.1

- Single unified Dart codebase delivering 99% code sharing across iOS (iPhone/iPad) and Android.
- Offline-first local caching using Hive / SQLite with automatic background synchronization.
- Strict adherence to Apple Human Interface Guidelines (HIG) and Material Design 3 standards.

5. DETAILED FUNCTIONAL SCOPE — CUSTOMER MOBILE APPLICATION

5.1 Onboarding & Multi-Factor Authentication

- Splash screen, interactive feature walkthrough, and language selection (English / Hindi / Arabic).
- Mobile OTP login (Twilio / MSG91), Email/Password login, Google Sign-In, and Sign in with Apple.
- Biometric quick-unlock (FaceID / TouchID) and secure token refresh via Flutter Secure Storage.

5.2 Home Dashboard, Catalog & Smart Search

- Dynamic banner carousels, personalized category feeds, and real-time promotional offers.
- Debounced full-text search with multi-attribute filters (price, rating, distance, availability).
- Deep linking (Universal Links / App Links) for direct sharing of items and referral campaigns.

5.3 Booking / Order Workflow & Payment Gateway

- Interactive cart & slot scheduling with coupon code validation and tax breakdown.
- Native SDK integration with Razorpay / Stripe supporting UPI, Cards, NetBanking, and Wallets.
- Automated PDF invoice generation, in-app order tracking, and instant refund status updates.

5.4 Real-Time Notifications & In-App Support Chat

- Rich push notifications via Firebase Cloud Messaging (FCM) with action buttons.
- Real-time WebSocket / Firestore customer support chat with image & document attachments.
- Ratings, verified reviews, and automated feedback prompts after order completion.

6. WEB ADMIN PANEL & ANALYTICS CONSOLE

- Executive KPI Dashboard: Daily Active Users (DAU), revenue charts, cohort retention, and conversion funnel.
- Role-Based Access Control (RBAC): Super Admin, Operations Manager, Finance Team, and Support Executive.
- Catalog & Content Management: Dynamic banners, service categories, pricing tiers, and promo codes.
- Settlement & Reports: Automated CSV/XLSX financial exports, GST/VAT reports, and audit logs.

7. SYSTEM ARCHITECTURE & FLUTTER 3.X ENGINEERING STACK

Our architecture follows Clean Architecture principles separating Presentation (Flutter Widgets + BLoC), Domain (Use Cases + Entities), and Data Layers (Repositories + Remote/Local Data Sources).

Architecture Tier	Selected Technologies & Standards
Presentation Layer	Flutter 3.x, Material 3 / Cupertino Widgets, BLoC Pattern, GoRouter
Local Persistence	Hive Encrypted Box, SQLite (Drift), Flutter Secure Keychain/Keystore
API Gateway & Auth	RESTful JSON APIs + WebSockets, OAuth 2.0 Bearer JWT, Rate Limiting
Application Server	Laravel 11 / Node.js Cluster, Horizon Queue Workers, Redis Cache
Cloud Infrastructure	AWS EC2 Auto-Scaling, RDS Multi-AZ PostgreSQL, S3 + CloudFront CDN
DevOps & Monitoring	GitHub Actions CI/CD, Fastlane, Firebase Crashlytics & Sentry APM

8. SECURITY, PERFORMANCE & COMPLIANCE BENCHMARKS

- TLS 1.3 SSL pinning to prevent Man-in-the-Middle (MitM) packet interception on public Wi-Fi.
- OWASP Mobile Top 10 compliance with ProGuard / R8 obfuscation for Android APK/AAB and iOS IPA.
- Sub-200ms API response time SLA for 95th percentile queries using Redis read-through caching.
- 60 FPS / 120 FPS Impeller rendering engine optimization with zero jank on frame rasterization.
- Automated daily encrypted database snapshots with 30-day point-in-time recovery (PITR).

9. DEDICATED PROJECT TEAM & RESOURCE ALLOCATION

Role / Designation	Allocation	Primary Responsibility
Project Manager / Scrum Master	**1 Dedicated**	Sprint planning, client demos, risk tracking & delivery
Senior Flutter Lead & Engineers	**2 Full-Time**	iOS/Android app architecture, BLoC state, native plugins
Senior Backend & Cloud Architect	**1 Full-Time**	REST APIs, DB schema, payment gateway & AWS setup
UI/UX Product Designer	**1 Shared**	Figma design system, interactive prototypes & assets
QA Automation & Manual Engineer	**1 Full-Time**	Device matrix testing, API Postman suite & regression

10. COMMERCIAL INVESTMENT & PAYMENT MILESTONES

Milestone	Deliverable Stage	Payment Tranche	Schedule
Milestone 1	Project Kickoff & Architecture Sign-Off	20% of Total Project Value	Upon Contract Signing
Milestone 2	UI/UX Wireframes & Interactive Figma Prototype	20% of Total Project Value	End of Week 2
Milestone 3	Alpha Release (Core Flutter Modules + Backend APIs)	25% of Total Project Value	End of Week 5
Milestone 4	Beta Release (Full Feature Complete + Admin Panel)	20% of Total Project Value	End of Week 8
Milestone 5	UAT Sign-Off, App Store / Play Store Live & Handover	15% of Total Project Value	End of Week 11

11. THIRD-PARTY INFRASTRUCTURE NOTES

- Apple Developer Program ($99/year) and Google Play Console ($25 one-time) accounts to be owned by Client.
- Cloud hosting (AWS / DigitalOcean) and SMS/Email gateway usage billed directly at actuals.

12. QUALITY ASSURANCE, WARRANTY & SLA SUPPORT

- 90 Days of complimentary post-launch warranty covering any functional bugs or regressions.
- Severity-1 (Critical Production Outage): Response within 1 hour, resolution within 4 hours.
- Severity-2 (Major Feature Degradation): Response within 4 hours, resolution within 12 hours.
- Comprehensive Testing: Unit tests, Flutter widget tests, integration tests, and load testing up to 10k concurrent users.

13. INTELLECTUAL PROPERTY (IP) & SOURCE CODE HANDOVER

- Upon final milestone completion, 100% of custom source code, Figma design files, database schemas, and API documentation become the exclusive intellectual property of the Client.
- Ajath Infotech transfers full Git repository ownership along with build pipelines and deployment keys.

14. PROPOSAL ACCEPTANCE & AUTHORIZED SIGN-OFF

By signing below, both parties agree to the scope, schedule, and terms set forth in Proposal v2.0.1:

FOR AJATH INFOTECH PVT. LTD.

Name: Manjot Singh
Title: Global VP & Partner
Date: January 30, 2023
Signature: Manjot Singh (Signed)

FOR CLIENT (AUTHORIZED SIGNATORY)

Name: _____
Title: _____
Date: _____
Signature: _____